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## 1. 习题三:

30.

(1) 由题意得

$$f(y) = \begin{cases} 2y, & 0 < y < 1 \\ 0, & \text{others} \end{cases}$$

因为 X, Y 相互独立, 则 (X, Y) 的联合概率密度为

$$f(x, y) = f_X(x)f_Y(y) = \begin{cases} 4xy, & 0 < x < 1, 0 < y < 1 \\ 0, & \text{others} \end{cases}$$

$$(2) P(Y > 2X) = \int_0^{\frac{1}{2}} \int_{2x}^1 f(x, y) dy dx = \frac{1}{8}$$

$$(3) f_Z(z) = \int_{-\infty}^{+\infty} f(x, z-x) dx = \begin{cases} \int_0^z 2x \cdot 2(z-x) dx, & 0 < z < 1 \\ \int_{z-1}^1 2x \cdot 2(z-x) dx & 1 < z < 2 \\ 0, & \text{others} \end{cases}$$

$$\therefore f_Z(z) = \begin{cases} \frac{2z^3}{3}, & 0 < z < 1 \\ 4z - \frac{2}{3}z^3 - \frac{8}{3}, & 1 < z < 2 \\ 0, & \text{others} \end{cases}$$

32.

$$f_Z(z) = \int_{-\infty}^{\infty} f(x, 2x-z) dx, 0 < x < 1, 0 < 2x-z < 2x$$

$$\text{即 } f_Z(z) = \int_{-\infty}^{\infty} f(x, 2x-z) dx = \begin{cases} \int_{\frac{z}{2}}^1 dx = 1 - \frac{z}{2}, & 0 < z < 2 \\ 0, & \text{others} \end{cases}$$

34.

由题意得, Y 的概率密度为

$$f_Y(y) = \begin{cases} \frac{10}{y^2}, & y > 10 \\ 0, & y \leq 10 \end{cases}$$

$$f_Z(z) = \int_{-\infty}^{+\infty} f_X(yz)f_Y(y)|y|dy = \begin{cases} \int_{\frac{10}{z}}^{+\infty} \frac{100}{y^3 z^2} dy = \frac{1}{2}, & 0 < z < 1 \\ \int_{10}^{+\infty} \frac{100}{y^3 z^2} dy = \frac{1}{2z^2}, & z \geq 1 \\ 0, & \text{others} \end{cases}$$

**37.**

由题意得

$$f_X(x) = \begin{cases} \frac{1}{2}, & 1 < x < 3 \\ 0, & \text{others} \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{2}, & 1 < y < 3 \\ 0, & \text{others} \end{cases}$$

$$\text{故 } f(x, y) = f_X(x)f_Y(y) = \begin{cases} \frac{1}{4}, & 1 < x < 3, 1 < y < 3 \\ 0, & \text{others} \end{cases}$$

$$F_Z(z) = P(Z \leq z) = P(|X-Y| \leq z) = \iint_{-z \leq x-y \leq z} f(x, y) dx dy = \begin{cases} 0, & z \leq 0 \\ 1 - \int_1^{3-z} dx \int_{x+z}^3 f(x, y) dy - \int_{z+1}^3 dx \int_1^{x-z} f(x, y) dy, & 0 < z < 2 \\ 1, & z \geq 2 \end{cases}$$

$$\therefore f_Z(z) = \begin{cases} 1 - \frac{z}{2}, & 0 < z < 2 \\ 0, & \text{others} \end{cases}$$

## 2. 习题四:

$$1. X \text{ 的可能取值为 } 1, 2, 3, 4 \quad P(X=1) = \frac{(1+C_3^2 \times 3 + C_3^1 \times 3^2)}{4^3} = \frac{37}{64}$$

$$P(X=2) = \frac{(1+C_3^2 \times 2 + C_3^1 \times 2^2)}{4^3} = \frac{19}{64}$$

$$P(X=3) = \frac{1+C_3^2+C_3^1}{4^3} = \frac{7}{64}$$

$$P(X=4) = \frac{1}{4^3} = \frac{1}{64}$$

$$E(X) = 1 \times P(X = 1) + 2 \times P(X = 2) + 3 \times P(X = 3) + 4 \times P(X = 4) = \frac{25}{16}$$

3

4. 令  $X =$  “首次检测到正品前检测到的次品数”， $X$  的可能取值为 0, 1, 2, 3

$$P(X = 0) = \frac{9}{12} = \frac{3}{4}$$

$$P(X = 1) = \frac{3}{12} \times \frac{9}{11} = \frac{9}{44}$$

$$P(X = 2) = \frac{3}{12} \times \frac{2}{11} \times \frac{9}{10} = \frac{9}{220}$$

$$P(X = 3) = \frac{3}{12} \times \frac{2}{11} \times \frac{1}{10} = \frac{1}{220}$$

$$E(x) = 0 \times P(X = 0) + 1 \times P(X = 1) + 2 \times P(X = 2) + 3 \times P(X = 3) = \frac{3}{10}$$

5. 令  $X =$  “电梯需停次数”，

$$X_i = \begin{cases} 0, & i \\ 1, & i \end{cases}, \quad X = \sum_{i=2}^n X_i$$

$$P(X_i = 0) = \left(\frac{n-2}{n-1}\right)^m, P(X_i = 1) = 1 - P(X_i = 0) = 1 - \left(\frac{n-2}{n-1}\right)^m$$

$$E(X_i) = 0 \times P(X_i = 0) + 1 \times P(X_i = 1) = 1 - \left(\frac{n-2}{n-1}\right)^m$$

$$E(X) = \sum_{i=2}^n E(X_i) = (n-1)\left(1 - \left(\frac{n-2}{n-1}\right)^m\right)$$

10.

$$\text{由题意得 } \int_{-\infty}^{\infty} f(x)dx = \int_{-1}^1 k(1-x^2)dx = \frac{4}{3}k = 1$$

$$\therefore k = \frac{3}{4}$$

$$E((X+500)^2) = \int_{-1}^1 \frac{3}{4}(x+500)^2(1-x^2)dx = 250000.2m^2$$

13.

$$\text{由题意得 } f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2}}$$

$$E(T) = -1 \times P(X < 10) + 20 \times P(10 \leq X \leq 12) + (-5) \times P(X > 12) = \frac{1}{\sqrt{2\pi}} \left( \int_{-\infty}^{10} -e^{-\frac{(x-\mu)^2}{2}} dx + \right.$$

$$4 \quad \int_{10}^{12} 20e^{-\frac{(x-\mu)^2}{2}} dx - \int_{12}^{+\infty} 5e^{-\frac{(x-\mu)^2}{2}} dx) \\ \frac{dE(T)}{d\mu} = \frac{1}{\sqrt{2\pi}}(-\int_{-\infty}^{10} (x-\mu)e^{-\frac{(x-\mu)^2}{2}} dx + 20 \int_{10}^{12} (x-\mu)e^{-\frac{(x-\mu)^2}{2}} dx - 5 \int_{12}^{+\infty} (x-\mu)e^{-\frac{(x-\mu)^2}{2}} dx) = \frac{1}{\sqrt{2\pi}}(21e^{-\frac{(10-\mu)^2}{2}} - 25e^{-\frac{(12-\mu)^2}{2}})$$

$$\text{令 } \frac{dE(T)}{d\mu} = 0, \text{ 解得 } \mu = \frac{1}{2}(\ln \frac{21}{25} + 22)$$

故当  $\mu = \frac{1}{2}(\ln \frac{21}{25} + 22) \approx 10.91mm$  时, 销售一个零件的平均利润最大

### 3. 3.4-6 补充题一:

(1) 由题意得

$$F_X(x) = \begin{cases} \int_0^x f_X(t)dt = 1 - e^{-\lambda x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

$$F_Y(y) = \begin{cases} \int_0^y f_Y(t)dt = 1 - e^{-\lambda y}, & y > 0 \\ 0, & y \leq 0 \end{cases}$$

$$F_M(m) = F_X(m)F_Y(m) = \begin{cases} (1 - e^{-\lambda m}), & m > 0 \\ 0, & m \leq 0 \end{cases}$$

$$F_N(n) = 1 - [1 - F_X(n)][1 - F_Y(n)] = \begin{cases} 1 - e^{-2\lambda n}, & n > 0 \\ 0, & n \leq 0 \end{cases}$$

$$P(M \leq u, N > v) = P(\max\{X, Y\} \leq u, \min\{X, Y\} > v) = \int_v^u \int_v^u f(x, y)dx dy = \int_v^u \int_v^u f_X(x)f_Y(y)dx dy = (e^{-\lambda u} - e^{-\lambda v})^2$$

$$(2) f(m, n) = \begin{cases} 0, & m < n \\ P(X = m, Y = n), & m = n \\ P(X = m, Y = n) + P(X = n, Y = m), & m > n \end{cases}$$

$$f(m, n) = \begin{cases} 0, & m < n \\ \lambda^2 e^{-2\lambda m}, & m = n \\ 2\lambda^2 e^{-\lambda(m+n)}, & m > n \end{cases}$$

## 4.1-6 补充题:

(1) 由题意得  $I_1, I_2$  的分布相同,

$$P(I_1 = 1) = P(X_1 > X) = \frac{15}{36} = \frac{5}{12}$$

$$P(I_1 = 0) = 1 - P(X_1 > X) = \frac{7}{12}$$

故  $(I_1, I_2)$  的联合分布列为

| $p_{ij}$                     |   | $I_1$            |                  | $p_{\cdot j} = \sum_i p_{ij}$ |
|------------------------------|---|------------------|------------------|-------------------------------|
|                              |   | 0                | 1                |                               |
| $I_2$                        | 0 | $\frac{49}{144}$ | $\frac{35}{144}$ | $\frac{7}{12}$                |
|                              | 1 | $\frac{35}{144}$ | $\frac{25}{144}$ | $\frac{5}{12}$                |
| $p_{i\cdot} = \sum_j p_{ij}$ |   | $\frac{7}{12}$   | $\frac{5}{12}$   | 1                             |

(2)  $\because I_1, I_2$  相互独立

$$\therefore E(I_1 I_2) = E(I_1)E(I_2) = E^2(I_1) = (1 \times P(I_1 = 1) + 0 \times P(I_1 = 0)) = \frac{25}{144}$$

$$(3) E((I_1 + I_2)^2) = 4 \times P(I_1 = 1, I_2 = 1) + 1 \times P(I_1 = 1, I_2 = 0) + 1 \times P(I_1 = 0, I_2 = 1) + 0 \times P(I_1 = 0, I_2 = 0) = \frac{85}{72}$$